

1 Model

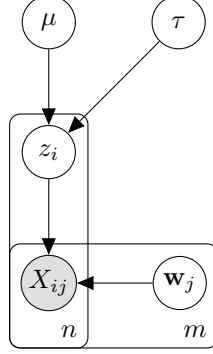


Figure 1: Bayesian network representing the latent regression model

Let $\mathbf{z} = \{z_1, \dots, z_n\}$ be a vector of independent and identically distributed Gaussian latent variables:

$$p(\mathbf{z}) = \prod_{i=1}^n \mathcal{N}(z_i \mid \mu, \tau^{-1}).$$

Let $\{\phi_1(\cdot), \dots, \phi_k(\cdot)\}$ denote a collection of basis functions. We define the vector-valued feature map

$$\phi(z_i) = (\phi_1(z_i), \dots, \phi_k(z_i))^\top \in \mathbb{R}^k.$$

The design matrix $Z \in \mathbb{R}^{n \times k}$ is then defined by stacking these vectors row-wise, so that

$$Z_{i\ell} = \phi_\ell(z_i).$$

Let $W \in \mathbb{R}^{k \times m}$ be a weight matrix, and $X \in \mathbb{R}^{n \times m}$ denote the observed data matrix. Conditioned on $\mathbf{z}$ and W , the entries of X are independent Gaussian random variables:

$$p(X \mid \mathbf{z}, W) = \prod_{i=1}^n \prod_{j=1}^m \mathcal{N}(X_{ij} \mid (ZW)_{ij}, \beta^{-1}),$$

where β is a known precision parameter.

Thus, conditioned on $\mathbf{z}$, the columns of X are independent and each column $\mathbf{x}_j$ corresponds to a separate linear regression model with its own weight vector $\mathbf{w}_j$.

We assume a noninformative prior over the latent variable parameters:

$$p(\mu, \tau) \propto \tau^{-1},$$

and place independent Gaussian priors on the columns of W :

$$\mathbf{w}_j \sim \mathcal{N}(0, \alpha_j^{-1} I), \quad j = 1, \dots, m,$$

where $\alpha_j > 0$ is a known precision hyperparameter associated with column j .

In order to formulate a variational treatment of this model, we consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\mathbf{z}, W, \mu, \tau) = q(\mathbf{z})q(W, \mu, \tau).$$

First, let us consider the derivation of the update equation for the factor $q(W, \mu, \tau)$. The log of the optimized factor is given by

$$\ln q^*(W, \mu, \tau) = \mathbb{E}_{q(\mathbf{z})} [\ln p(W, \mu, \tau \mid \mathbf{z}, X)] + \text{const.}$$

As can be seen from Figure 1, the d-separation criterion implies that

$$\mathbb{E}_{q(\mathbf{z})} [\ln p(W, \mu, \tau \mid \mathbf{z}, X)] = \mathbb{E}_{q(\mathbf{z})} [\ln p(\mu, \tau \mid \mathbf{z})] + \sum_j \mathbb{E}_{q(\mathbf{z})} [\ln p(\mathbf{w}_j \mid \mathbf{z}, X)].$$

Therefore, we have the following induced factorization for $q(W, \mu, \tau)$:

$$q(W, \mu, \tau) = q(\mu, \tau) \prod_{j=1}^m q(\mathbf{w}_j).$$

We obtain $q(\mathbf{w}_j)$ using the fact that each $\mathbf{w}_j$ is a weight vector of a bayesian linear regression model:

$$\begin{aligned} \ln p(\mathbf{w}_j \mid \mathbf{z}, X) &= -\frac{\beta}{2} \|Z\mathbf{w}_j - \mathbf{x}_j\|^2 - \frac{\alpha_j}{2} \mathbf{w}_j^\top \mathbf{w}_j + \text{const} \\ &= -\frac{1}{2} \mathbf{w}_j^\top (\beta Z^\top Z + \alpha_j I) \mathbf{w}_j + \beta \mathbf{x}_j^\top Z \mathbf{w}_j + \text{const}. \end{aligned}$$

Taking the expectation with respect to $q(\mathbf{z})$, we obtain

$$\ln q^*(\mathbf{w}_j) = -\frac{1}{2} \mathbf{w}_j^\top (\beta \mathbb{E}_{q(\mathbf{z})} [Z^\top Z] + \alpha_j I) \mathbf{w}_j + \beta \mathbf{x}_j^\top \mathbb{E}_{q(\mathbf{z})} [Z] \mathbf{w}_j + \text{const.}$$

The expression above is quadratic in $\mathbf{w}_j$. Since the logarithm of a multivariate Gaussian distribution is a quadratic function of its argument, it follows that

$$q^*(\mathbf{w}_j) = \mathcal{N}(\mathbf{w}_j \mid \mu_j, \Sigma_j),$$

We therefore have

$$\begin{aligned}\mathbb{E}[\mathbf{w}_j] &= \mu_j, \\ \mathbb{E}[\mathbf{w}_j \mathbf{w}_j^\top] &= \Sigma_j + \mu_j \mu_j^\top,\end{aligned}$$

where

$$\begin{aligned}\Sigma_j^{-1} &= \beta \mathbb{E}_{q(\mathbf{z})} [Z^\top Z] + \alpha_j I, \\ \mu_j &= \beta \Sigma_j \mathbb{E}_{q(\mathbf{z})} [Z]^\top \mathbf{x}_j.\end{aligned}$$

We now consider the update equation for the factor $q(\mu, \tau)$:

$$\begin{aligned}\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\mathbf{z})} [\ln p(\mu, \tau \mid \mathbf{z})] + \text{const} \\ &= -\frac{\tau}{2} \left(\sum_i \mathbb{E}_{q(z_i)} [z_i^2] - 2\mu \sum_i \mathbb{E}_{q(z_i)} [z_i] + n\mu^2 \right) \\ &\quad + \left(\frac{n}{2} - 1 \right) \ln \tau + \text{const}.\end{aligned}$$

We define

$$\begin{aligned}\bar{z} &= \frac{1}{n} \sum_i \mathbb{E}_{q(z_i)} [z_i], \\ \bar{z}\bar{z} &= \frac{1}{n} \sum_i \mathbb{E}_{q(z_i)} [z_i^2].\end{aligned}$$

We observe that $q^*(\mu, \tau)$, for $n > 1$ follows a Normal–Inverse–Gamma distribution:

$$q^*(\mu, \tau) = \mathcal{N} \left(\mu \mid \bar{z}, \frac{1}{n\tau} \right) \text{Gamma} \left(\tau \mid \frac{n-1}{2}, \frac{n(\bar{z}\bar{z} - \bar{z}^2)}{2} \right).$$

This implies

$$\begin{aligned}\mathbb{E}_{q(\tau)} [\tau] &= \frac{(n-1)}{n(\bar{z}\bar{z} - \bar{z}^2)}, \\ \mathbb{E}_{q(\mu, \tau)} [\mu\tau] &= \bar{z} \mathbb{E}_{q(\tau)} [\tau].\end{aligned}$$

Finally, we derive the update equation for $q(\mathbf{z})$:

$$\begin{aligned}\ln q^*(\mathbf{z}) &= \mathbb{E}_{q(W, \mu, \tau)} [\ln p(X \mid \mathbf{z}, W) + \ln p(\mathbf{z} \mid \mu, \tau)] + \text{const} \\ &= \mathbb{E}_{q(W)} [\ln p(X \mid \mathbf{z}, W)] + \mathbb{E}_{q(\mu, \tau)} [\ln p(\mathbf{z} \mid \mu, \tau)] + \text{const}.\end{aligned}$$

We expand the terms $\ln p(X \mid \mathbf{z}, W)$ and $\ln p(\mathbf{z} \mid \mu, \tau)$. All expressions that are independent of $\mathbf{z}$ are treated as constants.

$$\begin{aligned} \ln p(X \mid \mathbf{z}, W) &= \sum_j \ln \mathcal{N}(\mathbf{x}_j \mid Z\mathbf{w}_j, \beta^{-1}I) \\ &= -\frac{\beta}{2} \sum_j \|Z\mathbf{w}_j - \mathbf{x}_j\|^2 + \text{const} \\ &= \beta \sum_i \left(-\frac{1}{2} \phi(z_i)^\top \sum_j \mathbf{w}_j \mathbf{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j X_{ij} \mathbf{w}_j \right) + \text{const}. \end{aligned}$$

$$\begin{aligned} \ln p(\mathbf{z} \mid \mu, \tau) &= -\frac{\tau}{2} \sum_i (z_i - \mu)^2 + \text{const} \\ &= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau \mu z_i \right) + \text{const}. \end{aligned}$$

Substituting these expansions into the variational update expression yields:

$$\ln q^*(\mathbf{z}) = \sum_i \ln q^*(z_i),$$

where

$$\begin{aligned} \ln q^*(z_i) &= \beta \phi(z_i)^\top \sum_i \left(-\frac{1}{2} \sum_j \mathbb{E}_{q(\mathbf{w}_j)} [\mathbf{w}_j \mathbf{w}_j^\top] \phi(z_i) + \sum_j X_{ij} \mathbb{E}_{q(\mathbf{w}_j)} [\mathbf{w}_j] \right) \\ &\quad + \left(-\frac{1}{2} \mathbb{E}_{q(\tau)} [\tau] z_i^2 + \mathbb{E}_{q(\mu, \tau)} [\tau \mu] z_i \right) + \text{const}. \end{aligned}$$

This implies that $q(\mathbf{z})$ factorizes over its components z_i . The update equations therefore require expectations of the form

$$\mathbb{E}_{q(z_k)} [\phi_i(z_k) \phi_j(z_k)], \quad \mathbb{E}_{q(z_k)} [\phi_i(z_k)], \quad \mathbb{E}_{q(z_k)} [z_k], \quad \mathbb{E}_{q(z_k)} [z_k^2],$$

for all relevant indices i, j, k . These expectations can be evaluated numerically.